

Stimulus Design for Visual Evoked Potential Based Brain-Computer Interfaces

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Abstract—Visual stimuli design plays an important role in brain-computer interfaces (BCIs) based on visual evoked potentials (VEPs). Variations in stimulus parameters have been shown to affect both decoding accuracy and subjective perception experience, implying the need for a trade-off in design. In this study, we comprehensively and systematically compared various combinations of amplitude contrast and spectral content parameters in the stimulus design to quantify their impact on decoding performance and subject comfort. Specifically, three parameters were investigated: 1) contrast level, 2) temporal pattern (periodic steady-state or pseudo-random code-modulated), and 3) frequency range. We collected electroencephalogram (EEG) data and subjective perception ratings from ten subjects and evaluated the decoding accuracy and subject comfort rating for different combinations of the stimulus parameters. Our results indicate that while high-frequency steady-state VEP (SSVEP) stimuli were rated the most comfortable, they also had the lowest decoding accuracy. Conversely, low-frequency SSVEP stimuli were rated the least comfortable but had the highest decoding accuracy. Standard and high-frequency M-sequence code-modulated VEPs (c-VEPs) produced intermediates between the two. We observed a consistent trade-off relationship between decoding accuracy and subjective comfort level across all parameters. Based on our findings, we offer c-VEP as a preferable stimulus for achieving reliable decoding accuracy while maintaining a reasonable level of comfortability.

Index Terms—Brain-computer interfaces, electroencephalography, m-sequence, stimulus design, visual evoked potentials.

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I. INTRODUCTION

VISUAL evoked potential (VEP)-based brain-computer interface (BCI) has enabled a wide range of clinical and commercial applications [1], [2]. Examples include a BCI speller for locked-in patients to communicate with family and medical staff, integration with virtual-reality (VR) headset for visual field loss assessment for Glaucoma patients [3], and integration with VR and augmented-reality (AR) as additional control and capability for gaming [4], [5]. When compared to other BCI approaches, the VEP-based BCI has two advantages: (1) high information transfer rate (ITR) [6], [7], and (2) robust brain responses with limited human variability.

In a VEP-based BCI, a user gazes at one of multiple targets with visual stimuli each tagged with distinct properties, such as different frequencies and phases of periodic stimuli, or different pseudo-random sequences of coded stimuli, resulting in VEPs exhibiting a corresponding response specific to the target stimulus. The target therefore can be identified by analyzing the VEPs against a dictionary of target stimuli, and the command corresponding to the target can thus be transferred to an external device as a control signal. There have been various multiple-stimulus coding methods developed to generate distinguishable VEPs focused on achieving high classification accuracy and bandwidth. For example, steady-state VEPs (SSVEPs), which are modulated by periodic sequences at given frequencies and/or phases, have achieved ITRs of over 300 bits/min in previous studies [6], [7]. Likewise, code-modulated VEPs (c-VEPs) with broadband visual stimuli modulated by pseudo-random sequences have been targeted for high-speed BCI applications [8]. Prior research has shown that c-VEP-based BCIs can attain remarkable classification accuracy, leading to high ITRs [9], [10], [11], [12]. For example, a recent study used c-VEP to create a 120-command BCI with an ITR of 260 bits/min [13].

VEP-based BCI research has mainly focused on optimizing decoding accuracy and often overlooked an important human factor in the stimulus design—comfort, which is critical for user adoption and real-world applications. The key research questions investigated in this study are: (1) how to design more comfortable visual stimuli; (2) the relationship between comfort and decoding accuracy of a VEP-based BCI; and (3) how to jointly optimize parameters of visual stimulus design,

such as stimulus patterns, contrast levels, and frequencies, to achieve optimal trade-off between comfort and accuracy.

It is well known that subjective perceptual score decreases significantly as the flashing frequency of SSVEP stimuli is increased [14]. However, by modulating the luminance of the stimuli presentation, the perception could be restored to normal levels [15]. Recent studies have also shown that c-VEP offers higher ITR than SSVEP in VEP-based BCI settings [6]. At this time, there have been no reported studies directly comparing the comfort level of SSVEP and c-VEP. Likewise, no studies have been reported comparing the comfort level and decoding accuracy at different contrast levels and temporal patterns, whether for SSVEP or c-VEP.

This study aims to evaluate the effect of visual stimulus patterns and contrast levels on the subjective ratings of perception and comfort, and decoding accuracy from visual-evoked EEG responses. We designed and compared visual stimuli with combinations of three variables: (1) contrast levels; (2) temporal pattern (periodic steady-state versus pseudo-random code-modulated); and (3) stimulating frequency range, including a novel design of a high-frequency-modulated maximum-length sequence (HF-MSEQ) that has not been previously used in VEP-based BCI research. The study's findings could lead to the development of more comfortable visual stimuli for real-world BCI applications in the future.

II. METHODS

A. Visual Stimulus Design and Presentation

Visual stimuli were presented on a 27-inch LCD monitor (ASUS VG278) with a refresh rate of 60 Hz and a resolution of 1280×960 pixels (4:3 ratio). The screen was evenly divided into four quadrants to present four visual stimuli (targets) simultaneously. Each quadrant (640×480 pixels) consisted of a center area of 601×451 pixels that presented the visual stimulus and non-flickering separating boundaries with a width of 39 pixels horizontally and 29 pixels vertically. The separating boundaries were to reduce the spatial contrast between two adjacent visual stimuli.

A gray background (128 in a range of 0 to 255, where 0 and 255 are black and white, respectively) was used throughout the experiment. A visual stimulus was generated by up- or down-modulating the gray-scale intensity from the baseline for each 1/60-second frame according to a binary code sequence. This study tested three levels of intensity modulation (i.e. contrasts): 2, 8, and 16. For example, for a code sequence of "0, 1, 1, 0" with a contrast level of 2, the intensity levels of the four consecutive frames are "128-2, 128+2, 128+2, 128-2". These values were selected based on the result of a pilot study on five subjects who reported that a contrast level of 2 was barely perceivable.

This study used four types of multiple-stimulus coding: low-frequency SSVEP, high-frequency SSVEP, maximum-length sequence c-VEP, and high-frequency modulated maximum-length sequence c-VEP. The design of binary code sequences generated by the four methods is further described in the following sections.

1) *Steady-State Visual-Evoked Potential (SSVEP)*: The code sequence for inducing SSVEP has a single frequency component in the frequency domain and is oscillatory in the

time domain. A sinusoidal sequence was first generated by: $s(f, i) = \sin(2\pi f(i/R))$, where i is the frame index in the sequence, R is the refresh rate (i.e., 60Hz in this study) of the monitor, and f denotes the frequency of the sequence [16], [17], [18]. The sinusoidal sequence was converted to the final binary sequence based on whether the value is greater (1/+) or less (0/-) than 0. For low-frequency SSVEP (LF-SSVEP), we used $f = 8, 8.67, 9.33$, and 10 Hz, for the four targets on the screen of the monitor, respectively. For high-frequency SSVEP (HF-SSVEP), $f = 24, 26, 28$, and 30 Hz were used. Note that 30Hz is the highest (Nyquist) frequency achievable with most conventional devices having a 60Hz refresh rate. These frequencies were selected such that LF-SSVEP and HF-SSVEP have the same resolution (e.g. $(10 - 9.33)/10 = (30 - 28)/30 = 6.7\%$), allowing for a fair comparison for target identification.

2) *Maximum-Length Sequence (MSEQ)*: A maximum-length sequence is a pseudo-random binary sequence generated by a maximum-length linear feedback shift registers (LFSR) with a random except all-zeros initial state. An 8-bit LFSR was used to generate $2^8 - 1 = 255$ -bit long binary sequences (with feedback polynomial $x^8 + x^6 + x^5 + x^4 + 1$) [19], and the first 180 bits (3 seconds) of the sequences were used for the experiment. Four sequences were generated with different initial states and being orthogonal to each other. MSEQ has a uniform distribution in the frequency domain, giving a broad-band stimulation that visually resembles the candle-light effect, and has been used for VEP-based brain-computer interfaces [8]. This type of VEP and its extensions to other broadband modulation sequences are called *code-modulated VEP* (c-VEP).

3) *High-Frequency Modulated Maximum-Length Sequence (HF-MSEQ)*: We designed a new binary code sequence for c-VEP that is based on MSEQ but with fewer low-frequency components than MSEQ. Fig. 1 depicts the concept of this method. Having a binary code sequence generated by MSEQ (e.g., 1 0 1 0 0 1) whose frequency components are uniformly distributed between 0 and 30 Hz at a frame rate of 60 Hz, we first interpolated the sequence by duplicating each code value (e.g., 1 0 1 0 0 1 becomes 11 00 11 00 00 11). Because this process was equivalent to doubling the temporal resolution, the frequency components of the sequence became uniformly distributed between 0 and 15 Hz. We then convolved the interpolated MSEQ with a high-frequency (30 Hz) carrier by flipping every other bit in the sequence (e.g. 11 00 11 00 00 11 becomes 10 01 10 01 01 10), resulting in a broad-band activity between 15 to 30 Hz and reduced signals at low frequency—between 0 to 15 Hz. The high-frequency-modulated MSEQ (HF-MSEQ) offers the advantage that it is free from low-frequency content, and hence is thought to be less perceptible and more comfortable than the conventional MSEQ. Furthermore, its exclusively high-frequency modulated content allows low-frequency content to be superimposed with minimal interference, supporting BCI applications with dynamic targets or with streaming video up to a 30 Hz frame rate (15 Hz bandwidth).

Fig. 2 shows examples of the LF-SSVEP, MSEQ, HF-MSEQ, and HF-SSVEP stimulus sequences used in the experiments, and the corresponding VEP responses obtained by

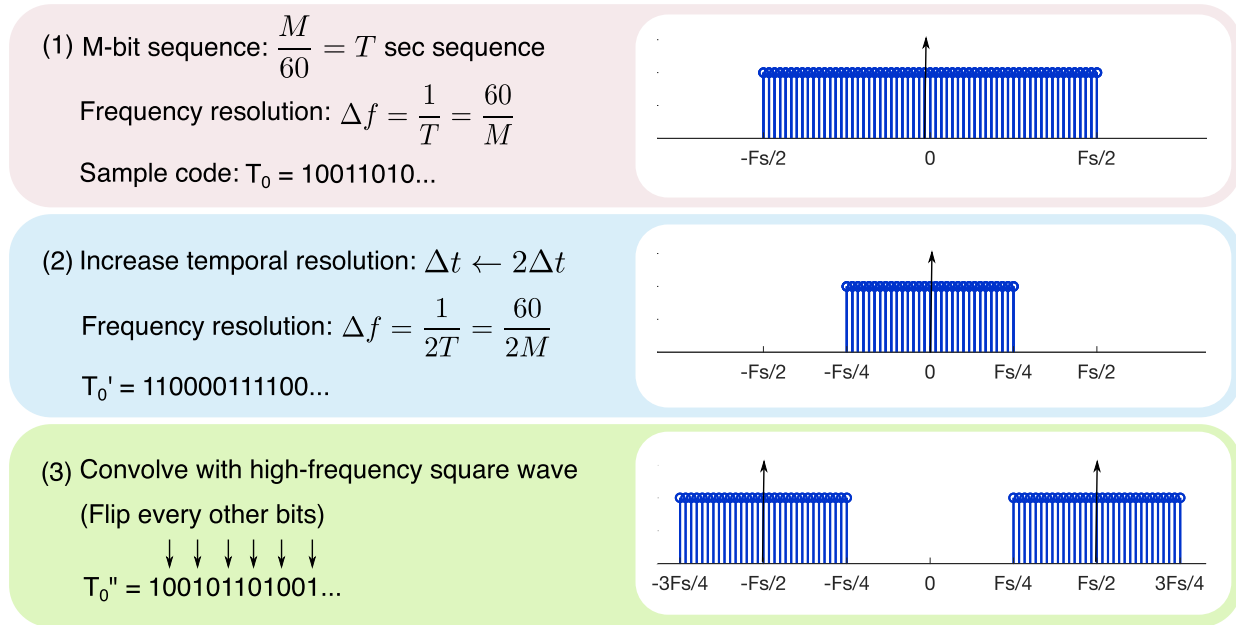


Fig. 1. A schematic illustration of the process for generating a high-frequency maximum-length sequence (HF-MSEQ).

averaging 10 recording trials at channel Oz. The amplitude spectra for SSVEP stimuli showed clear fundamental (i.e., 10 Hz for LF-SSVEP and 30 Hz for HF-SSVEP) and/or harmonic frequencies (e.g., 20 Hz for LF-SSVEP) as shown in Fig. 2(i) and (l). In contrast, c-VEP sequences such as MSEQ (Fig. 2(j)) and HF-MSEQ (Fig. 2(k)) resulted in a broad-frequency response without any clear single-peaked components. It demonstrated that the four stimulus sequences successfully induced distinct frequency responses from each other.

B. Experiments

Ten healthy subjects (5 female, age: 20.6 ± 2.46 years) with normal or corrected-to-normal vision participated in this experiment after signing the consent form approved by the Human Research Protections Program of the University of California San Diego. The subjects were instructed to sit in a dimly lit room and 60 cm in front of the monitor. The experiment consists of five sessions separated by self-paced breaks. Each session consisted of 8 blocks, each with 13 trials, resulting in a total of 520 trials. One training block was given to the subjects to familiarize themselves with the experiment. Each trial began with the presentation of a black fixation cross at the center of one target among the four sections on the screen of the monitor (See Section II-A for details), followed by a 3-second stimulation period. The subjects were instructed to shift their gaze to the target, fixate at the cross throughout the stimulation period, and avoid movement and eye blinks during the stimulation. At the end of the trial, subjects rated their perceived intensity of the stimuli and comfortability (Table I). In pilot testing, we separated the perceived intensity and comfortability of stimuli into two questions and found that the two ratings were highly inversely correlated. As a result, in the experiments, we combined them into a single question. The entire experiment took approximately 60 minutes for each subject.

The stimuli at each trial were swept over two variables: *contrast level* (2, 8, and 16) and *stimulus type* (LF-SSVEP,

TABLE I
QUESTIONNAIRE ON PERCEIVED INTENSITY AND COMFORTABILITY OF THE VISUAL STIMULI

Score	Description
1	Not Perceptible
2	Barely Perceptible / Not Annoying
3	Perceptible / Slightly Annoying
4	Quite Strong Stimuli / Annoying
5	Strong Stimuli / Very Annoying
6	Very Strong Stimuli / Uncomfortable
7	Extremely Strong Stimuli / Very Uncomfortable

HF-SSVEP, MSEQ, and HF-MSEQ), with a total of $3 \times 4 = 12$ conditions. We also added another condition using LF-SSVEP with contrast of 16 superimposed with static images rather than a gray background to motivate the subjects and maintain their attention in the experiment. These trials were excluded from the data analysis in this study. The 13 conditions were repeated ten times for each of the four target locations on the screen, resulting in $13 \times 4 \times 10 = 520$ trials. The contrast levels, stimulus types, and target locations were all randomized across the 520 trials for each subject. It is worth noting that, for each trial, the contrast level and the type of stimuli were identical for the four visual stimuli on the screen, with only their frequencies (for HF-SSVEP and LF-SSVEP) or their initial states (for MSEQ and HF-MSEQ) differing. The design and the presentation of the visual stimuli and the experimental paradigm were implemented in MATLAB (The Mathworks, Inc) using the Psychophysics Toolbox [20].

C. EEG Acquisition and Preprocessing

EEG data were recorded by 32 Ag/AgCl electrodes covering the parietal and occipital areas using a BioSemi ActiveTwo EEG system at a sampling rate of 512 Hz. Two reference electrodes were placed at the mastoids. The EEG data were high-pass filtered at 1 Hz with a zero-phase forward and reverse filter using the `filtfilt()` function in MATLAB. The `cleanline()` plugin in EEGLAB was used to remove

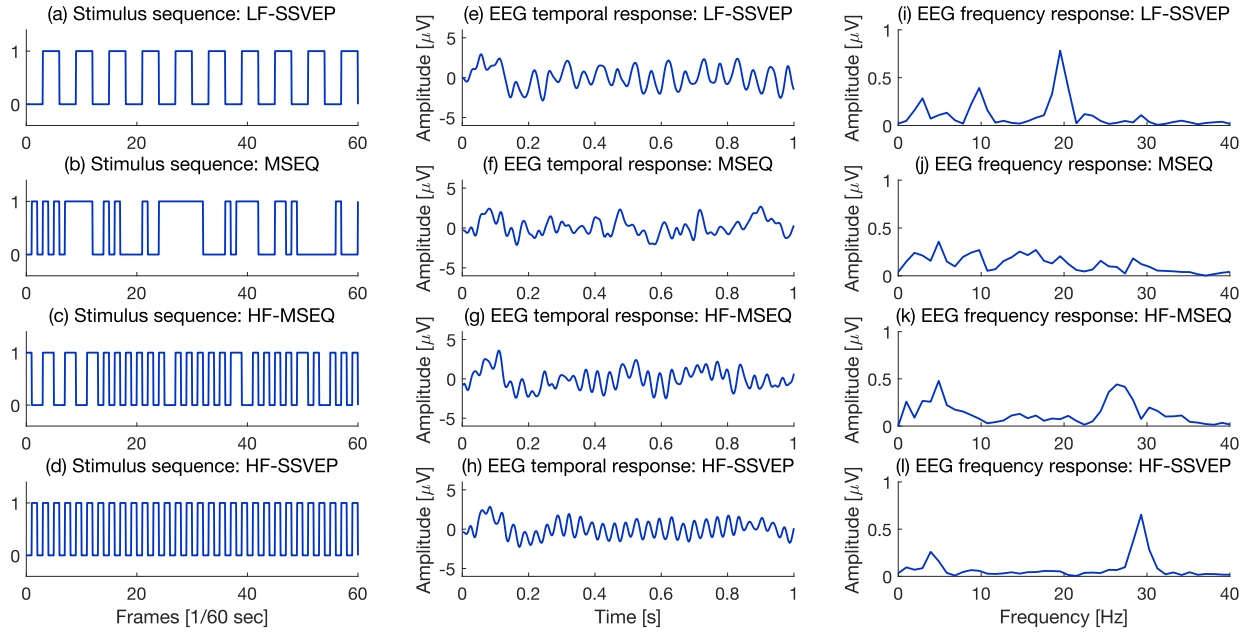


Fig. 2. Examples of 1-second (60 frames) code sequences of four types of stimuli (a)-(d) and their corresponding visual-evoked EEG activities at channel Oz averaged across all trials and subjects in the time domain (e)-(h) and frequency domain (i)-(l).

60-Hz line noise. Artifact subspace reconstruction (ASR) with a cutoff parameter 20 [21] was used to remove large-amplitude artifacts and reconstruct clean signals. The data epochs were extracted in $[0.1 \text{ sec}, 0.1+d \text{ sec}]$, where time 0 represents the stimulus onset and d the data length.

D. EEG Analysis

We used the accuracy of the target identification to quantitatively compare the decodability of EEG responses to visual stimuli with different stimulus types and contrasts. This study employed the template-based target identification method [22], [23] with the ensemble task-related component analysis (TRCA)-based spatial filtering method [7]. The template-based approach has been widely used in detecting both SSVEP and c-VEP [7], [8]. In addition, a previous comparison study of various spatial-filtering techniques verified that TRCA-based spatial filtering was the most effective approach in enhancing the signal-to-noise ratio (SNR) of evoked potentials including SSVEPs [24].

1) **Target Identification:** TRCA is the method that extracts task-related components efficiently by maximizing their reproducibility during task periods [25]. Considering observed multi-channel EEG signals $\mathbf{x}(t) \in \mathbb{R}^{N_c}$, TRCA finds a linear projection coefficient vector $\mathbf{w} \in \mathbb{R}^{N_c}$ to maximize inter-trial correlation of task-related components $y(t) = \mathbf{w}^T \mathbf{x}(t)$. The h -th trial in the observed EEG signals and task-related components are described as $\mathbf{x}^{(h)} \in \mathbb{R}^{N_c \times N_s}$ and $\mathbf{y}^{(h)} \in \mathbb{R}^{N_s}$, respectively. The covariance C_{h_1, h_2} between h_1 -th and h_2 -th trials of $\mathbf{y}^{(h)}$ is described as:

$$\begin{aligned} C_{h_1, h_2} &= \text{Cov}(\mathbf{y}^{(h_1)}, \mathbf{y}^{(h_2)}) \\ &= \sum_{j_1, j_2=1}^{N_c} w_{j_1} w_{j_2} \text{Cov}(\mathbf{x}_{j_1}^{(h_1)}, \mathbf{x}_{j_2}^{(h_2)}). \end{aligned} \quad (1)$$

All possible distinct inter-trial combinations over N_t trials are summed as:

$$\begin{aligned} \sum_{\substack{h_1, h_2=1 \\ h_1 \neq h_2}}^{N_t} C_{h_1, h_2} &= \sum_{\substack{h_1, h_2=1 \\ h_1 \neq h_2}}^{N_t} \sum_{j_1, j_2=1}^{N_c} w_{j_1} w_{j_2} \text{Cov}(\mathbf{x}_{j_1}^{(h_1)}, \mathbf{x}_{j_2}^{(h_2)}) \\ &= \mathbf{w}^T \mathbf{S} \mathbf{w} \end{aligned} \quad (2)$$

where the inter-trial signal covariance matrix $\mathbf{S} \in \mathbb{R}^{N_c \times N_c}$ is defined as:

$$(S)_{j_1, j_2} = \sum_{\substack{h_1, h_2=1 \\ h_1 \neq h_2}}^{N_t} \text{Cov}(\mathbf{x}_{j_1}^{(h_1)}, \mathbf{x}_{j_2}^{(h_2)}). \quad (3)$$

To normalize the solution, the variance of $y(t)$ is constrained as:

$$\begin{aligned} \text{Var}(y(t)) &= \sum_{j_1, j_2=1}^{N_c} w_{j_1} w_{j_2} \text{Cov}(\mathbf{x}_{j_1}(t), \mathbf{x}_{j_2}(t)) \\ &= \mathbf{w}^T \mathbf{Q} \mathbf{w} \\ &= 1 \end{aligned} \quad (4)$$

where $\mathbf{Q} \in \mathbb{R}^{N_c \times N_c}$ is the covariance of the signal concatenated over all trials, $\mathbf{x}(t) = \mathbf{x}^{(1)} \mathbf{x}^{(2)} \dots \mathbf{x}^{(N_t)}$. The constrained optimization problem can then be solved as:

$$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\text{argmax}} \frac{\mathbf{w}^T \mathbf{S} \mathbf{w}}{\mathbf{w}^T \mathbf{Q} \mathbf{w}}. \quad (5)$$

The optimal coefficient vector $\hat{\mathbf{w}}$ is obtained from eigenanalysis of the matrix $\mathbf{Q}^{-1} \mathbf{S}$, selecting the eigenvector corresponding to the largest eigenvalue as the spatial filter to extract task-related components.

For VEP-based BCI, TRCA can be used to design spatial filters tuned to VEP responses specific to each stimulus, while removing background EEG activity from the scalp recordings. Spatial filters $\mathbf{w}_n \in \mathbb{R}^{N_c}$ for each stimulus $n = 1, \dots, N_f$

(frequency or phase for SSVEP, or code for c-VEP) are obtained by TRCA from individual calibration data $\mathbf{x}_n \in \mathbb{R}^{N_c \times N_s \times N_t}$. The correlation coefficient between single-trial test data $\mathbf{x} \in \mathbb{R}^{N_c \times N_s}$ and an individual template for the n -th visual stimulus $\bar{\mathbf{x}}_n$ is calculated as follows:

$$r_n = \rho(\mathbf{w}_n^T \mathbf{x}, \mathbf{w}_n^T \bar{\mathbf{x}}_n), \quad (6)$$

where the stimulus templates are obtained by averaging the calibration data over trials, $\bar{\mathbf{x}}_n = \frac{1}{N_t} \sum_h \mathbf{x}_n^{(h)}$. The target stimulus τ is then identified by finding the one that maximizes correlation across all stimuli as $\tau = \operatorname{argmax}_n r_n$.

2) Performance Evaluation and Statistical Analysis: The recorded EEG epochs were classified by the template-based method with the TRCA-based spatial filter described in the previous section. The classification accuracy was estimated by leave-one-trial-out cross-validation, in which all combinations of nine and one trials were used for training and testing, respectively, for each visual stimulus design, and for each individual subject. We evaluated classification performance with varying data lengths, d , ranging from 0.1 to 3 s with a 0.1 s interval.

The evaluation of classification performance as well as subjective ratings on the intensity and comfortability of the presented stimuli were submitted to two-way repeated measures analysis of variance (ANOVA) for all four types of stimuli (LF-SSVEP, HF-SSVEP, MSEQ, and HF-MSEQ), times all three levels of contrast (2, 8, and 16). In addition, the averaged classification accuracy across different data lengths d was also submitted to two-way repeated measures ANOVA with four types of stimuli, times three contrasts. In post-hoc analysis, the subjective ratings and the decoding accuracy were compared between pairs of different stimulation types at each contrast level using t -test with multiple comparison corrections. Because we compared 18 pairs ($3! = 6$ pairs at each of the three contrast levels), the significance level was corrected to 0.0028 (i.e., $0.05 / 18$). This study employed `statcond()` in EEGLAB [26] to perform the factorial repeated measures ANOVAs and the post-hoc t -tests.

III. RESULTS

A. Subjective Ratings of Perceived Intensity and Comfortability

Fig. 3 shows the averaged subjective ratings across subjects for each stimulation type and contrast level. The results show that the low contrast levels and the high-frequency stimulations (i.e., HF-SSVEP and HF-MSEQ) lead to better comfortability than the others. In fact, two-way repeated measures ANOVA showed main effects of contrasts ($F(2, 27) = 235.95$; $p < 0.001$) and types of stimuli ($F(3, 27) = 22.50$; $p < 0.001$), but no significant interaction between them ($F(6, 54) = 1.05$; $p = 0.402$). Post-hoc analysis using paired t -tests with Bonferroni correction showed that the perception scores for HF-SSVEP were significantly lower than the other three types of stimuli at the contrast levels of 2 (vs. HF-MSEQ: $p = 0.0001$; vs. MSEQ: $p = 0.000$; vs. LF-SSVEP: $p = 0.0002$) and 8 (vs. HF-MSEQ: $p = 0.0010$; vs. MSEQ: $p = 0.0006$; vs. LF-SSVEP: $p = 0.0010$). No significant difference was present between types of stimuli at contrast

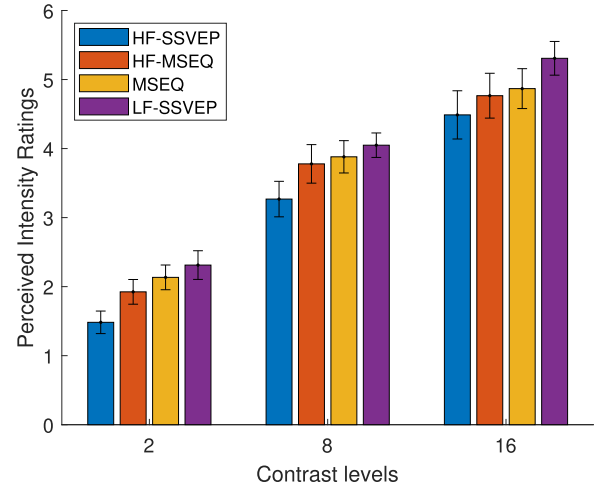


Fig. 3. Mean and standard errors of subjective rating of perceived intensity of the four stimuli types at three contrast levels.

level 16, nor between MSEQ and LF-SSVEP at any of the contrast levels.

B. Decoding Performance of Visual Stimuli

Fig. 4 shows the averaged decoding accuracy across subjects with different data lengths at contrast levels of 2, 8, and 16 in relation to different stimulus types. At contrast 2, only LF-SSVEP with data lengths greater than 0.3 seconds and LF-MSEQ with data lengths greater than 0.1 seconds performed significantly better accuracy than chance (i.e. 25%). At contrast 8, most of the stimulus patterns, with the exception of HF-SSVEP, showed decoding accuracy over 50% when the data length was greater than 0.3 seconds. At contrast 16, the decoding accuracy was greatly improved for all the stimulus patterns. The classification accuracy generally increased with longer data lengths especially when the data length was shorter than 500 milliseconds. However, longer data did not significantly improve accuracy in situations where accuracy was close to chance level (e.g., HF-SSVEP at contrast level 2) or had almost reached 100% accuracy (e.g., LF-SSVEP at contrast level 16).

Fig. 5 shows the averaged decoding accuracy across data lengths and subjects at each contrast level and stimulation pattern. Two-way repeated measures ANOVA found main effects on contrast ($F(2, 27) = 101.35$; $p < 0.001$) and types of stimuli ($F(3, 27) = 31.98$; $p < 0.001$). No interaction was found between contrast and stimulus types ($F(6, 54) = 1.56$; $p = 0.200$). Post-hoc analysis using paired t -tests with Bonferroni correction showed that the decoding accuracy for HF-SSVEP was significantly lower than the other three types of stimuli at the contrast levels of 8 (vs. HF-MSEQ: $p = 0.0018$; vs. MSEQ: $p = 0.0008$; vs. LF-SSVEP: $p = 0.0015$). At the contrast level of 16, the decoding accuracy of HF-SSVEP was significantly lower than that of HF-MSEQ ($p = 0.0020$). No significant difference was found in the other comparisons.

IV. DISCUSSION

In this study, we aimed to test different combinations of parameters for creating comfortable yet decodable visual stimuli for VEP-based BCIs. Many studies had previously

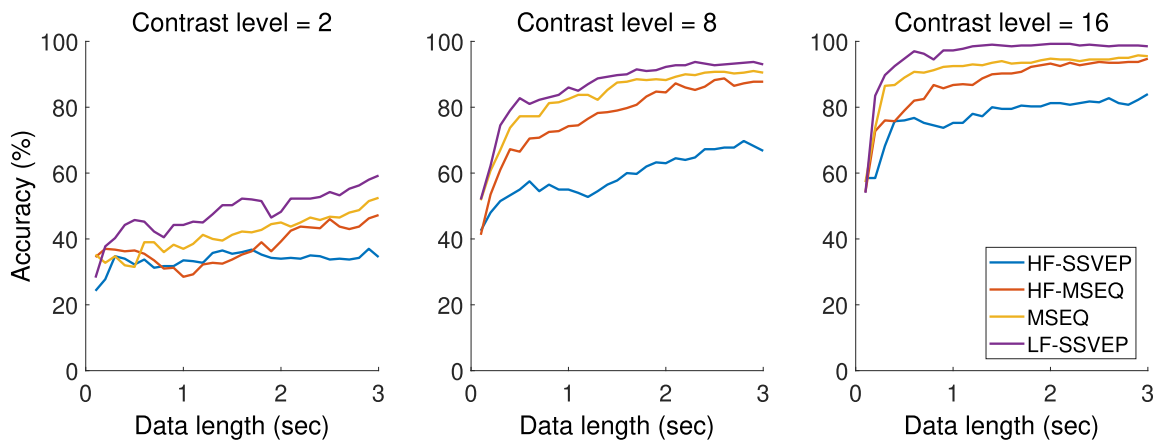


Fig. 4. Cross-validation accuracy across subjects with different training data lengths using TRCA for each type of stimuli at contrast levels 2, 8, and 16.

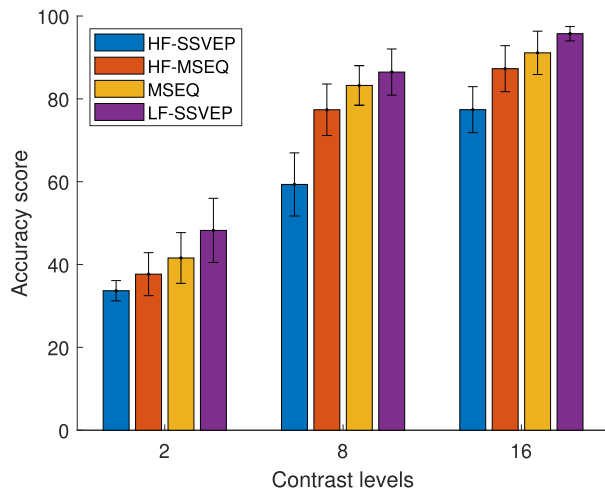


Fig. 5. Cross-validation accuracy across subjects with four stimuli at contrast levels 2, 8, and 16.

tested and compared the decoding accuracy of SSVEPs elicited by various stimulation methods. Yet the combined performance of both comfortability and decodability has remained unclear. Thus, this study compared the comfortability and decodability of three parameters: frequencies, stimulus patterns, and contrast levels. We were particularly interested in the performance of the high-frequency M-sequence (HF-MSEQ). In theory, the higher carrier frequency (30Hz) of HF-MSEQ should be less perceptible, potentially reducing visual fatigue.

Across all contrast and pattern conditions, our results indicated that high-frequency stimuli rendered a lower perception rating than low-frequency stimuli (Fig. 3). This is consistent with previous research that the perception threshold increases as the frequency increases [14]. In contrast, the contrast level rendered a reversed trend, with the perception threshold decreasing as the contrast level increased [15]. Our results combined both factors and took a step further to explore perception ratings in relation to various patterns. Across all contrast levels, there was a consistent pattern that HF-SSVEP received the lowest perception score and LF-SSVEP received the highest. This implies that, in terms of comfort, HF-SSVEP is the best option to choose. MSEQ is a good middle-ground option, while LF-SSVEP is the worst option.

For decoding accuracy, there was an obvious trend that the decoding accuracy increased as the contrast level increased

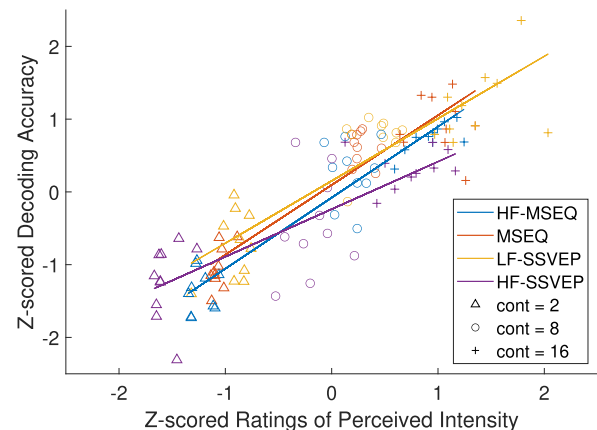


Fig. 6. The relationship between decoding accuracy and perception ratings. Each marker represents the normalized decoding accuracy and rating score obtained from the individual subjects for each stimulation pattern and contrast level.

(Fig. 4 and Fig. 5). Similar to the perception rating, the frequency level followed a reversed order where decoding accuracy decreased as the frequency increased. Despite having the lowest perception score, HF-SSVEP also had the lowest decoding accuracy among various stimulus patterns. For other stimulus patterns, LF-SSVEP achieved the highest decoding accuracy followed by MSEQ and HF-MSEQ. It is worth noting that this ranking in decoding accuracy is exactly the reverse of the ranking in comfortability.

Our results revealed that the stimulation method that could achieve a better perception experience (less intense stimuli) leads to lower decoding accuracy. Fig. 6 illustrates the relationship between the subjective ratings and classification accuracy for each contrast level and stimulation pattern. Regardless of the stimulus pattern, the stimuli with higher perceived intensity always produced a better decoding accuracy. This finding is consistent with the results of the previous study [27], which concluded that HF-SSVEP can alleviate visual fatigue whereas the decoding accuracy needs to be sacrificed. Therefore, the trade-off between comfortability and decodability is inevitable. Our result further confirms that c-VEP stimuli also follow the same trend. However, although not statistically significant, HF-SSVEP seems to show the weakest trade-off in comparison to all other stimulus patterns, indicating that it cannot improve the decoding accuracy as much as others even though the comfortability is comparable.

This study used TRCA, a data-driven classification method based on templates, to report decoding accuracy, enabling a fair comparison across different stimulus designs. TRCA is also the most popular algorithm for SSVEP- and c-VEP-based BCIs [7], [28]. Nevertheless, it is important to note that an algorithm optimized for each specific stimulus pattern could potentially achieve higher decoding accuracy. In particular, for code-based stimuli such as MSEQ and HF-MSEQ, the classification algorithm should leverage the available information from the known stimulus sequence (i.e., input) to further enhance decoding accuracy. If successful, MSEQ and HF-MSEQ could be preferred options for VEP-based BCIs, as they offer enhanced user comfort and sufficient accuracy.

This study focused on optimizing visual stimuli design for practical use in VEP-based BCIs. It did not comprehensively investigate the effects of EEG electrode placement and number on decoding performance. This study collected EEG data using electrodes covering only the parietal and occipital areas, where the VEPs are typically most prominent. Expanding the spatial coverage of electrodes could enable the exploration of optimal electrode placement and the minimum number of electrodes needed for satisfactory decoding performance. This could also enhance user convenience and comfort in the design of VEP-based BCIs.

V. CONCLUSION

This study conducted a comparison study to examine the effect of stimulus pattern, contrast level, and flashing frequency on decoding accuracy and subjective perception/comfort level in VEP-based BCIs. We found that LF-SSVEP had the best decoding accuracy among all the stimulus patterns, regardless of contrast level, while also having the lowest comfort score. In contrast, HF-SSVEP had the lowest decoding accuracy across all contrast levels, while also having the highest comfort score. The result suggested that there is an inevitable trade-off relationship between decoding accuracy and subjective comfort level. MSEQ and HF-MSEQ appear to be a preferable intermediate choice as a balance point on both sides of the trade-off, although the consistency of the relationship with approximately negatively linear dependence implies that no single point along the trade-off axis offers a distinct advantage with superior accuracy and comfort.

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